Categorial Features and The Feature Structure of the Present Tense: Perspectives from Variation and Change*

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Abstract: This paper investigates the syntax of verbless sentences in Arabic in the present tense, contending that the key to understanding the syntax of finite verbless sentences resides in an analysis of the nature of the present tense in Arabic varieties and particularly its feature specifications along the lines of Chomsky (1995) and Benmamoun (2000). Many Arabic varieties lack a copula in present tense sentences with non-verbal predicates, while past tense and future sentences require one. According to Benmamoun (2000), this property and a number of other morphosyntactic asymmetries between the present tense and other tenses in Arabic varieties can be explained under a view that they differ in their categorial feature specifications: while the past tense, for example, is specified as both [+V] and [+D], the present tense is only specified for the [+D] feature, making crucial use of the categorial features of Chomsky (1995). Unlike the majority of Arabic varieties, Sason Arabic has an obligatory copula in the present tense as well. We argue that the variation and change in the copular clauses is not an isolated phenomenon, but reflects a deeper change in the overall feature structure of the present tense. The present tense in Sason Arabic is specified as [+V,+D], and the development of an overt copula and other facts naturally follow from this shift, i.e., the emergence of the [+V] feature. In general, this proposal takes seriously the view that the functional domain of the clause and its categorial features drive syntactic dependencies, and also syntactic variation and change (e.g., Borer 1984).

Keywords: categorial features; tense; copula; language change; Arabic

1 Introduction

Since the inception of generative grammar in the second half of the last century, one perennial issue has been the components of clause structure, and the relationships and dependencies between those components. This is understandable given the central role clause structure plays in capturing key syntactic patterns and generalizations within all theories of generative grammar that rely on hierarchical relations and dependencies with the clause as a major unit of syntactic analysis. Those generalizations include anaphoric dependencies, agreement and agreement asymmetries, unbounded dependencies (such as questions and relatives), semantic scope, ellipsis, etc. The common thread that runs through all the approaches over the last 70 years is that members of the syntactic phrase marker occupy designated positions and enter into dependencies with each other, which in turn explains why a phrase (and its head) may have a different interpretation and surface realization depending on where it is in the phrase marker and what other member(s) of the phrase it is in relation with.

Thus, the clause and the hierarchical relations between its members (including nominals) are a fundamental concept that is a bedrock of all configurational approaches to syntactic patterns and generalizations. However, while there is consensus about the role of clause structure, there has been a vigorous debate about the internal structure of the clause and its members. For example, it is not very controversial that some dependencies are sensitive to the features on the lexical head and not to the semantic properties of the verb per se. The nominal case in many languages occurs in the context of finite tense with finiteness as the key feature. This could be captured by referencing finiteness on the verb or by referencing finiteness as an independent element. The latter has gained currency partly because there are good arguments that show that tense and finiteness are independent of the predicate. Thus, in English (1), the verb carries past tense, but in interrogative and negative sentences, a bleached verb "do" carries tense (Chomsky 1957):

(1) a. She wrote the letter.
    b. Did she write the letter?
    c. She didn't write the letter.

Standard Arabic provides a different type of evidence that support the same thesis. Here, it is negation that carries tense, not the verb, as shown in (2):

(2) a. katab-at r-risaalat-a
    write.PST-3F.SG the-letter-ACC
    'She wrote the letter.'
    b. lam ta-ktub r-risaalat-a
    NEG.PST 3F.SG-write the-letter-ACC
    'She didn't write the letter.' (Standard Arabic)

What the data in (1) and (2) show is that tense functions independently from the verb, and since nominative Case in English and Arabic is dependent on tense, it could then be reasonably argued that the latter occupies its own node within clause structure. The label for that node has changed significantly over the decades (Cinque 1999), but the fundamental assumption remains, namely that the clause has both a lexical layer (occupied by the predicate) and a functional layer that hosts features as tense.[1]

In addition to accounting for patterns and generalizations within individual languages, the functional layer has also been playing a critical role in accounting for syntactic variation between languages. For example, updating Chomsky's (1957) analysis of the distribution of tense inflection in English, Pollock 1989 relies on the functional category tense to account for variation in the positioning of lexical verbs and their direct objects relative to adverbs in English and French. Simplifying significantly, the main idea is that tense in French overtly causes the verb to move to tense past the adverb, but does not do so in English (at least for lexical verbs), hence the observed variation in word order. Pollock does not provide compelling arguments for why the functional layer, particularly tense, behave differently in the two languages. While there is a good case to be made that the verb gets displaced to a higher place, possibly Tense or even C, research is still ongoing about what motivates this displacement (see e.g., Emonds 1978; Vikner 1997; Rohrbacher 1999; Bobaljik 2002; Koeneman and Zeijlstra 2014). Nevertheless the main point that is relevant to us is the assumption that variation is to be captured through the workings of the functional layer, which echoes earlier work by Borer (1984).

The Minimalist Program took this idea even further reaching its culmination in Chomsky (1995) where the functional layer assumes an increasingly prominent role in accounting for syntactic patterns such as movement or lack thereof. Thus, assuming that the subject is generated within the lexical layer, the reason for the movement of the subject is due to the presence of a feature in the functional layer that requires pairing with (checking by) a nominal element (the subject). Verbs serve a similar function as they enter into dependencies with a verbal feature in the functional layer. Chomsky labels the two features as [+D] and [+V] categorial features, respectively. Essentially, what they indicate is that the functional layer drives syntactic dependencies that engage elements in the lexical layer, yielding movement patterns and unbounded dependencies and agreement at a distance.

The feature [±D] continues to play a central role in syntax including the syntax of functional projections beyond Tense, appearing in various versions (see e.g., Adger 2003; Müller 2010; Schäfer 2008, 2017 and many other works). For example, Schäfer 2017 proposes a Voice typology which is built on the [±D] feature. However, the role of the [±V] feature has not received similar attention, especially after the idea of categorial features was not given a central emphasis in Chomsky's later studies. The alternatives that have been proposed to replace categorial features have not overcome the same challenges, namely how to independently ground the features.[2] However, putting aside these differences, there seems to be a consensus that functional features drive syntactic patterns such as movement and syntactic dependencies in general. We will assume that this intuition that the functional layer drives key aspects of (narrow) syntax may be sound, and we will rely on this approach to account for intriguing and previously unaccounted variation in clause structure within Arabic varieties.

The main focus will be on copular constructions in the present and past tenses. In Arabic varieties, such as Moroccan Arabic (MA) and Standard Arabic, the verbal copula from the root /KWN/ is overt in the past tense, but not in the present tense, as shown in (3). By contrast, Sason Arabic (SA; an endangered Arabic variety spoken in southeastern Turkey) present tense sentences have an obligatory copula, (4b). Besides being head-final (see Akkuş and Benmamoun 2016, and section 3.2 for more discussion), the existing past forms are redeployed in the present tense. Therefore, the present copula is systematically homophonous/syncretic to the past tense copula, (4a). In (4) relevant adverbs are needed to disambiguate the temporal reference, as we will discuss in detail.

(3) a. ʔana kun-t f-d-dar.
I be.PST-1SG in-the-house
'I was in the house.'

b. ʔana f-d-dar.
I in-the-house
'I am in the house.' (Moroccan Arabic)

(4) a. ams fı beyt kın-tu.
yesterday in house be.PST-1SG
'I was in the house yesterday.'

b. sāhane fı beyt kın-tu.
now in house be.PRS-1SG
'I am in the house now.' (Sason Arabic)

This raises the question of why a verbal copula is required both in the present tense and in the past tense in SA, but not in MA and Standard Arabic. One option is that the variation in the present tense is simply a matter of surface realization, such that MA simply has a phonologically null counterpart of the SA copula. We will argue, instead, that the contrast between (3b) and (4b) reveals a deeper discrepancy in the clausal workings in these varieties. Specifically, the main factor has to do with the feature specifications of the present tense in MA and Standard Arabic (and other well-studied dialects), on one hand, and Sason Arabic, on the other. Adopting Chomsky's (1995) [+D] and [+V] features, Benmamoun (2000) demonstrates that in many Arabic varieties, the past and future tenses are specified for both of these features, whereas the present tense only for the [+D] feature even in sentences with verbal predicates.[3] This asymmetry captures a number of morpho-syntactic discrepancies between the two tenses in Arabic, some of which we discuss later in this paper too. We contend that the development of a copula in the present tense in SA is not an isolated phenomenon, but instead reflects a deeper change in the overall language, i.e., the feature composition of the tense system. Specifically, SA has undergone a change in which not just past tense, but also the present tense contains both [+D] and [+V] features, unlike MA and Standard Arabic. As such, the emergence of a copula is simply a natural consequence of this shift. On a higher level, this is consistent with the theory that the functional layer and its featural specifications play a critical role in narrow syntax and in accounting for variation both language internally (past vs. present tense) and also between languages (Borer 1984). We will argue that this is precisely what happened in the syntactic evolution of SA.

We close the paper with reflections on the possible role of contact with neighboring head-final languages such as Kurdish and Turkish in the emergence and surface position of the verbal copula in SA, as well as the potential diachronic trajectory as to what reanalysis and change(s) in parameter settings facilitated the current properties of clausal structure in SA. Additionally, we briefly mention other Anatolian Arabic varieties, which also evolved a copula in the present tense, but maintained the head-initial order, which shows the independence of the evolution of the verbal copula in the present tense from the changes in the word order.

2 Copula constructions and categorial features in Arabic varieties

In this section, we first introduce the verbless sentences from Moroccan Arabic and Standard Arabic, as representatives of many varieties, which distinguish between present and past tenses in terms of the possibility of a copula. Overviewing the analyses provided to capture this asymmetry, we adopt the approach in Benmamoun 2000, which connects the issue to the feature specifications of the tenses, where the present and past tense carry different features.

2.1 Copular constructions in Arabic varieties

In Moroccan Arabic (MA) and many Arabic varieties, a matrix sentence in the present tense can consist of only the subject and its lexical predicate as in (5), in contrast to past tense sentences which require a verbal element (to be shown in (11)).

(5) a. {ʔana / nta / ʕumar} muʕəllim.
I / you / Omar teacher
'{I am / you are / Omar is} a teacher.'

b. {ʔana / nta} ṭwil.
I / you tall
'{I am / you are} tall.'

c. d-dar kbira.
the-house big
'The house is big.'

d. {ʔana / nta / lə-ktab} f-d-dar.
I / you / the-book in-the-house
'{I am / you are / the book is} in the house.' (Moroccan Arabic)

Over the last six decades sentences such as (5), commonly referred to as *verbless sentences*, have received competing analyses from those that claim that they are matrix small clauses (consisting only of a lexical layer) to those who claim that they are full fledged clauses with a functional layer, which makes the sentences in (5) not too different from their English counterparts, except for the overt realization of the copula and its functional features.

As Benmamoun (2000) contended, there are strong arguments for adopting the analysis where the sentences in (5) are similar to their English counterparts in at least having a functional layer, and being different from the embedded genuine English small clauses, as in *I found [John angry]*. We will summarize a couple of the arguments in Benmamoun 2000 below.

One argument comes from Nominative case in Standard Arabic. The subject in Standard Arabic carries nominative in sentences with finite tense: in (6a), the indefinite subject is in nominative Case, which is usually associated with the subject of finite clauses with verbal predicates, as in (6b):

(6) a. fii l-bayt-i walad-u-n.
in the-house-GEN boy-NOM-INDF
'A boy is in the house.'

b. waqafa walad-u-n.
stand.PST.3M.SG boy-NOM-INDF
'A boy stood up.' (Standard Arabic)

Assuming that the Case on the indefinite subject in (6a) is not the default Case associated with topics, the simplest and plausible analysis is that its source of nominative Case as is the same as in (6b), namely finite tense, which does not have a phonological matrix (phonologically null). Indeed, positing a tense node separate from the predicate to account for sentences such as (6a) receives independent support from Standard Arabic sentences with expletive subjects, shown in (7):

(7) hunaaka walad-u-n fii l-bayt-i.
there boy-NOM-INDF in the-house-GEN
'There is a boy in the house.' (Standard Arabic)

It is widely accepted that expletive subjects such as *hunaaka* 'there' in (7) are located in the functional domain, while the lexical subject may be in the lexical domain (multiple subject positions). Assuming that this is correct, then in (7) one has to posit a functional node, plausibly headed by tense, that hosts the expletive subject. Benmamoun (2000) provides more arguments for an abstract functional layer for the verbless sentences in (5), but we will just add one more argument here and refer the reader to the original source. This argument is based on the dependency on between the complementizer domain and tense in Arabic varieties. Thus, in Moroccan Arabic, the complementizer *bəlli* requires a finite clause as complement, as in (8).

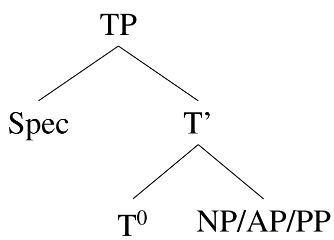
(8) sməʕt bəlli ʕumar ža.
hear.PST.1SG that Omar come.PST.3M.SG
'I heard that Omar came.' (Moroccan Arabic)

Likewise, verbless sentences can also be embedded under the complementizer *bəlli* as in (9), which bolsters the argument that verbless sentences contain a functional domain with tense features.

(9) sməʕt bəlli d-dar kbira.
hear.PST.1SG that the-house big
'I heard that the house is big.' (Moroccan Arabic)

Benmamoun (2000) took these pieces of evidence and others to argued that verbless sentences are not small clauses that contain only lexical projections, but full fledged clauses that contain a functional domain along the broad outline of the analysis of verbless sentences advanced by Doron for Hebrew (e.g., Doron 1983). The structure proposed by Benmamoun (2000) is reproduced in (10):

(10)



The structure in (10) provides a straightforward analysis for nominative Case, the expletive subject, and embedding under complementizers that require subordinate clauses. T^0 in (10) licenses nominative case, its specifier hosts the expletive subjects, and satisfies the selectional requirements of the C *bəlli*.

However, the immediate question that arises once one adopts the structure in (10) is the nature of tense head in copular constructions in the present tense. While there is no overt copula in the present tense, as in (5), a copula is required in the past tense and future tense. Consider (11).[4]

(11) a. kunt hna.
be.PST.1SG here
'I was here.'

b. l-wəld kan hna.
the-boy be.PST.3M.SG here
'The boy was here.'

c. ɣadi-n-kun hna.
FUT-1SG-be here
'I will be here.'

d. l-wəld ɣadi-y-kun hna.
the-boy FUT-3M.SG-be here
'The boy will be here.' (Moroccan Arabic)

It is tempting to generalize on the basis of (11) and posit a null verbal copula in copular constructions in the present tense. This would put them on par with their English counterparts, except for the overt realization of the copula and its functional features. Different versions of this analysis have been advanced over the last several decades (see e.g., Bakir 1980; Fassi Fehri 1993; Shlonsky 1997). However, there are strong arguments against the presence of a verbal copula in the structure of the Arabic sentences in (5) even though they involve a T head. Those arguments are discussed in detail in Benmamoun 2000, 2008, but here we will focus on the one based on Case.[5]

In Standard Arabic, the predicate that follows the verbal copula must be in accusative Case as in (12):

(12) a. kaana l-kitaab-u mufiid-an.
be.PST.3M.SG the-book-NOM interesting-ACC
'The book was interesting.'

b. sa-yakuunu l-kitaab-u mufiid-an.
FUT-be.3M.SG the-book-NOM interesting-ACC
'The book will be interesting.' (Standard Arabic)

Sentences in (12) indicate that there is a Case dependency between the copula and the adjectival predicate. Regardless of how Case dependencies are captured, the main point for us is that the presence of the copula determines the Case on the predicate. Interestingly, in sentences without an overt copula the predicate surfaces with nominative as in (13a), while accusative on the predicate is ruled out, (13b).

(13) a. al-kitaab-u mufiid-un.
the-book-NOM interesting-NOM
'The book is interesting.'

b. *al-kitaab-u mufiid-an.
the-book-NOM interesting-ACC
Intended: 'The book is interesting.'

Nominative Case in Standard Arabic is usually associated with subjects of tensed clauses and with topics and left dislocated elements that usually bind resumptive pronouns, in which case nominative is considered default. That analysis would extend straightforwardly to (13a), if we assume that there is no verbal head to license accusative Case on the predicate, which in turn leads to default nominative. Thus, based on Case in the context of the verbal copula, it would be reasonable that no such verbal copula exists in verbless present tense sentences. In other words, those sentences are truly verbless as their name implies.

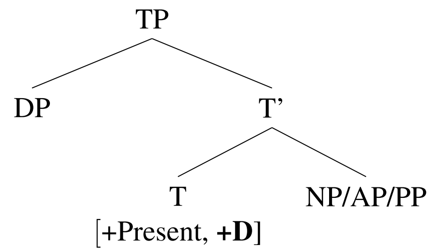
But if there is no verbal copula, then the structure of verbless sentences is as in (10) where Tense is projected to realize the time reference of the sentence and license the subject, including expletive subjects, and to satisfy the selectional requirements of the complementizer. The acute question that arises then is why there is a verbal copula in the past and future tense, but not in the present tense. Benmamoun (2000) ties the explanation to the feature composition of the tense adopting the 'categorial features' view in Chomsky 1995. According to Chomsky's (1995) theory, functional categories such as Tense can be specified for categorial features, [+D] and [+V], that in turn allows them to enter into (checking) relation with the subject and the verbal head. That relationship can be realized as movement and merger with TP (subject in Spec,TP and the verb adjoined to the T head). Chomsky further argued that those categorial features may trigger overt movement in some languages, but not in others, determined by the 'strength' of features (strong versus weak features), yielding the variation in the order of the finite verb and the adverb in French and English. While there have been attempts to independently motivate why some heads trigger overt movement while others may not, the arguments have been tentative and mainly speculative. It is not the purpose of this paper to take up that issue, but we believe that there is strong merit to the idea that tense is specified for categorial features (Cf. footnote 4), and that languages can differ in whether the present tense includes both features or only one. For the rest of the paper, we will assume that theory of categorial features to account for some fundamental differences between Arabic varieties.

The next section provides an initial illustration of the categorial features of tenses, and how it applies to many Arabic varieties. We leave a more in-depth elaboration to the later parts of the paper where we introduce Sason Arabic into the discussion.

2.2 Categorial features of Tense in Arabic varieties

The above system makes use of categorial features to derive parametric variation. Thus, according to the analysis developed in Benmamoun 2000, the present tense in Arabic is only specified for the nominal categorial feature [+D], and a verbless sentence in the present tense would have the structure in (14), where the subject DP moves to Spec,TP.

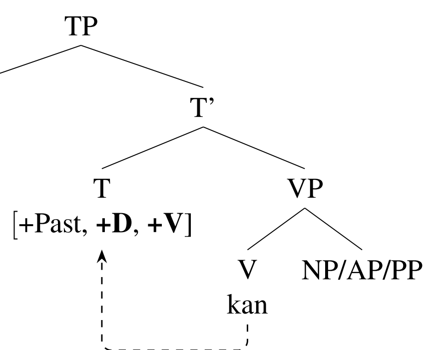
(14)



Since the present tense is only [+D] and does not carry the verbal feature [+V], it only interacts with the subject licensing its nominative Case and establishes a relationship between T and the external argument (EPP). In this respect Moroccan Arabic and Standard Arabic, which have the structure and feature specifications in (14), differ from English where present tense is both [+D] and [+V]. Thus, no verbal head is required in the present tense in Moroccan Arabic and Standard Arabic, hence verbless sentences.

On the other hand, the past and future tenses are both [+D] and [+V]. For example, the past tense sentence has the structure and feature specification in (15), hence also the verbal copula in the past as an overt reflex of the [+V] feature checking. This copula furthermore raises to the T head (see Fassi Fehri 1993; Benmamoun 2000; Ouali 2014; Jarrah and Abusalim 2020, a.o., for V-to-T raising of the copula).

(15)



As we will illustrate in the next section via comparisons between Arabic varieties, the categorial specifications of the tenses also hold in sentences with verbal predicates. Thus, although a VP is available in the clausal spine with verbal predicates in all tenses, T is still specified only for [+D] in the present tense,[6] while it is specified for [+D,+V] in the past/future tenses, as evidenced by a number of contrasts between the tenses including V-to-T movement in the past, but not present tense. We will take this up below in more details, but the critical point is that the feature specification of tense is uniform in present tense sentences, regardless whether the predicate is verbal or non-verbal.

The idea that the locus of variation with the feature specifications of the tense head in sentences with and without verbal copulas is consistent with the theory that many aspects of syntactic variation have to do with the workings of the functional domain of the clause, particularly the temporal and complementizer domains.[7] Thus, a particular functional head may have one categorial feature in one language and may totally lack it in another. In the same vein, functional categories within the same language may have different categorial feature specifications. For the languages under consideration in this paper, the difference between Moroccan Arabic and Standard Arabic on one hand, and English and French on the other has to do with the feature structure of the present tense. One advantage of this approach is that it allows us to capture the similarities between past and future tense sentences in the four languages and the differences between them in the present tense. That is because these four languages differ only in the categorial features of the present tense. If this analysis is on the right, then we predict that there could be Arabic varieties that differ from Moroccan Arabic and Standard Arabic but that pattern with French and English. In other words, we predict that there could be a variety of Arabic where the present tense is [+D] and [+V], which also affects the verbless sentences. Thus, we expect to receive the following combinations in (16):

(16) *Present tense and copula in Arabic varieties*

	Moroccan/Standard Arabic etc.	a hypothetical Arabic variety
Categorial features	[+D]	[+D, +V]
Copula	no	yes

We take this prediction up in the next section by discussing data that as far as know has not been discussed in this context before. We demonstrate that Sason Arabic is one such Arabic variety which has undergone a categorial change in its present tense specification. This shift in turn made possible the emergence of a verbal copula, exactly as the analysis predicts.

3 Changes in the categorial features

In this section, we discuss Sason Arabic, which exemplifies a situation in which a language undergoes a change in terms of the categorial features of the present tense. An immediate effect of this change is the development of an overt copula in the present tense, which behaves like the past tense.

3.1 Copula in Sason Arabic

Sason Arabic (SA) is one of several Arabic varieties spoken in Anatolia (Jastrow 1978, 2006; Akkuş 2017, 2020a), which are part of the so-called *qəltu*-dialect branch of the larger Mesopotamian Arabic.[8] It also fits in the category of a *peripheral* Arabic variety, as it is geographically spoken outside the 'Arab world'.

Although it does not have the robust Semitic root and pattern system, SA still maintains many of the Arabic characteristics in terms of its morphosyntax. For example, SA is, for the most part, a head-initial language like other Arabic varieties. The examples in (17) illustrate the head-initial nature of the language in various domains like CP, TP, VP and PP.

(17) a. mo-[a]-saddex [CP le zɣar [T kın-no [VP kı-i-qfıl-o atsura-ma]]].
NEG-1SG-believe that children be.PAST-3PL ASP-3-catch.PRS-PL bird-a
'I don't believe that the children were catching a bird.'

b. [PP ša Herdem]
for Herdem
'for Herdem' (Sason Arabic)

As in other Semitic languages and Arabic dialects (Aoun et al. 2010), verbs in SA exhibit two morphological patterns affecting the placement of its agreement morphology: perfective and imperfective. In the simple past tense, expressed using the perfective form of the verb, subject agreement is realized as a suffix on the verb, (18a). In the present tense (continuous or habitual), by contrast, which uses the imperfective, the realization of agreement differs dramatically. It is realized by both prefixes and suffixes, (18b).

(18) a. faqas-**te**
run.PST-2SG.F
'You (sg.f) ran.'

b. **tı**-fqız-**e**
2-run.PRS-F
'You (sg.f) run.' (Sason Arabic)

In other regards as well, SA clausal spine does not seem to be that different from other most-studied Arabic dialects (see Akkuş 2014, 2021a,b; Akkuş and Benmamoun 2018 for some studies on different syntactic aspects of SA, and Aoun et al. 2010 for an overview of the syntax of a few major Arabic dialects). For example, like other colloquial Arabic varieties, it lacks overt case and mood markings on nouns and verbs, respectively. The clause structure in SA consists of C > Neg > T > Asp > Voice > *v* > Root, as in other Arabic varieties, and in addition to having left periphery domain, it also host a low discourse-related area between TP and AspP including FocusP (similar to Jordanian Arabic (Jarrah and Abusalim 2020), or Lebanese Arabic (Ouwayda and Shlonsky 2016). In the interest of space, we do not exemplify these aspects of SA since they have been described and discussed in the above-cited studies.

Still SA exhibits certain patterns some of whose presence can perhaps be attributed to its contact with the head-final neighboring languages, Turkish, Kurdish, Zazaki and Armenian. Among these patterns are the weakening of the extent and the role of the templatic morphology peculiar to Semitic languages, the loss of the definite article, the use of periphrastic causative constructions rather than gemination or the "ablaut" causative, and the development of some head-final properties such as the light-verb construction (see Akkuş and Benmamoun 2018; Akkuş 2020a, 2024 for details).

While the absence of an overt copula in the present tense sentences is a well-known characteristic of the Arabic varieties spoken in the Arab world (cf. section 2), SA deviates from this pattern in that it has evolved a head-final copula in those same contexts where it is absent in other Arabic varieties. Consider (19), which illustrates this with different types of predicates and relevant temporal adverbs. Note that the use of copula in the present sentences is *obligatory*, and this copula agrees in person, number, and gender (we discuss in section 3.2 that the position of the copula in SA is also due to language contact).

(19) a. ina {raxu / zaki / muallim / nihane / fı beyt } *(kın-tu).
I {sick / intelligent / teacher / here / in house } be-1SG
'I am {sick / intelligent / a teacher / here / in the house}.'

b. {sāhane / lome} cuan-e *(kın-te).
{now / today} hungry-F be-2SG.F
'You(.sg.f) are sick {now / today}.'

c. ınto daima koys-in *(kın-to).
you.PL always beautiful-PL be-2PL
'You(.pl) are always beautiful.' (Sason Arabic)

Intriguingly, the present tense copula has drawn on the past tense paradigm, and is almost entirely identical to the past auxiliary, as shown in (20). Thus, the language appears to have redeployed existing forms to also express a new tense interpretation.[9]

(20) The copula paradigm in Sason Arabic tenses

	Present	**Past**
1sg	kıntu	kıntu
1pl	kınna	kınna
2m.sg.	kınt	kınt
2f.sg.	kınte	kınte
2f.pl.	kınto	kınto
3pl.	kınno	kınno, kano

Given the almost complete homophony, a nonverbal sentence without a context or an appropriate temporal adverb would be ambiguous between a present and past tense interpretation. Accordingly, all the examples in (19) can be used in the past episodic or generic contexts depending on the predicate, as illustrated in (21).

(21) a. ina {raxu / zaki / muallim / nihane / fı beyt } *(kın-tu).
I {sick / intelligent / teacher / here / in house } be-1SG
'I was {sick / intelligent / a teacher / here / in the house}.'

b. ams cuan-e *(kın-te).
yesterday hungry-F be-2SG.F
'You(.sg.f) were sick yesterday.'

c. ɪnto qɪddam koys-in *(kɪn-to).
you.PL before beautiful-PL be-2PL
'You(.pl) were beautiful in the past.' (Sason Arabic)

Note that temporal adverbs in (Sason) Arabic narrow down temporal reference in predictable way, and are well-behaved from a crosslinguistic perspective. Therefore, the language otherwise maintains a temporal distinction between past versus present tenses, (22).

(22) a. zɣer i-kri {lome / sāhane / ɣade / sɪnnare / *ams / *amlol }
child 3M-write.PRS {today / now / tomorrow / next year / *yesterday / last year }
maitub-ma.
letter-a
'The child writes / will write a letter {today / now / tomorrow / next year / *yesterday / *last year}.'

b. zɣer kara {ams / amlol / *ɣade / *sɪnnare } maitub-ma.
child write.PST.3M {yesterday / last year / *tomorrow / *next year } letter-a
'The child wrote a letter {yesterday / last year / *tomorrow / *next year}.'[10]

The copular construction in SA is informative with respect to the status of verbless sentences in Arabic. In particular, it argues against the approaches that (i) take verbless clauses to be essentially root small clauses with no functional projection (e.g., Mouchaweh 1986, Rapoport 1987), or (ii) assume that verbless sentences contain a verbal copula which is either null/zero (Fassi Fehri 1993) or subsequently erased (e.g., Bakir 1980). Importantly, SA has not resorted to a present tense form of the root KWN, which is what would one expect under the null (covert) verbal copula analysis for other dialects (see section 3.1.2 for discussion).

Instead, the SA copula fits straightforwardly into an approach which takes verbless clauses to have a reduced structure with some functional structure (Jelinek 1981; Doron 1983; Benmamoun 2000, 2008; Choueiri 2016). In this approach, it is not a question of whether the present tense copula is always verbal, but overt in some varieties and covert in others. Rather, SA created a new verbal copula in the present tense using cells from the past tense [+V, +D] paradigm. As noted above, this proposal follows from the view that the functional domain is the locus of variation and syntactic change (e.g., Borer 1984), such that we contend that the key to understanding verbless sentences resides in an analysis of the nature of the syntax of the various tenses in Arabic varieties. Accordingly, it is the change in the categorial features of tense that led to the development of the verbal copula in the present tense. Various pieces of evidence indicate that the present tense in Sason Arabic has adopted the [+V, +D] paradigm wholesale, thus is levelled in terms of the categorial features in the present and past tenses. As such, the copula in verbless clauses is simply an extension of this unified behavior of the tenses.

3.1.1 Change in the feature structure of present tense

Recall that Benmamoun (2000) concluded, based on various pieces of evidence, that the present tense in Arabic is only specified for the nominal categorial feature [+D], whereas the past/future are specified both for [+D] and [+V] in *non*-peripheral Arabic varieties. This is schematized in (23).

(23) Tense in non-peripheral Arabic varieties
  a. Past/future: [+V, +D]
  b. Present: [+D]

This characterization has been widely assumed and shown to capture many properties in a number of Arabic dialects including Egyptian Arabic, Moroccan Arabic, Hijazi Arabic, and Standard Arabic (see Aoun et al. 2010; Karawani and Zeijlstra 2013; Alqassas 2015; Jarrah 2023, a.m.o. for works that adopted this proposal, including for Palestinian Arabic and Jordanian Arabic). According to this analysis, the perfective is a tense operator in Arabic varieties occupying T, whereas bare imperfectives do not, and instead remain adjoined to *v* or move only up to Asp (see also Soltan 2007 for more evidence that perfective verb forms occupy a higher position than the imperfectives).

Various key facts regarding the position of arguments follow from this analysis, e.g., obligatory V-to-T in the past, but not in the present tense even in sentences with verbal predicates. This is borne out, as illustrated via various contrasts in Benmamoun 2000, 2008 (see also Fassi Fehri 1993 for similar discussion). One example comes from the so-called Ferguson's *God wishes*, realized with VS order in the perfective, (24a), and SV in the imperfective, (24b), even those with the same roots (see Benmamoun 2000; Soltan 2007 for more examples from quite a few varieties including Standard Arabic, Syrian Arabic).[11]

(24) a. barak llahu fii-k!
  bless.PST.3SG.M God in-you
  'May God bless you!'
 b. llah y-barik fii-k!
  God 3SG.M-bless.PRS in-you
  'May God bless you!' (MA)

Another example in the same vein comes from Jordanian Arabic. Jarrah (2023) demonstrates that in present tense clauses, the surface subject of the passive must appear preverbally, shown in (25) (the same contrast holds for indefinite subjects too; Marwan Jarrah, pers. comm.).

(25) a. (ʔis-sijja:rah) ʔib-ti-n-ħarig (*ʔis-sijja:rah).
the-car IPFV-3SG.F-PASS-burn the-car
'The car is (being) burned.'

b. (ʔis-siʕir) b-n-rafiʕ (*ʔis-siʕir).
the-price IPFV-PASS-raise.3SG.M the-price
'The price is (being) raised.' (present clauses in Jordanian Arabic (Jarrah 2023:14))

On the other hand, the surface subject in past passive clauses are allowed post-verbally too. Consider (26).

(26) a. (ʔis-sijja:rah) ʔin-ħarag-at (ʔis-sijja:rah).
the-car PASS.burn.PST-3SG.F the-car
'The car was burned.'

b. (ʔis-siʕir) ʔin-rafaʕ (ʔis-siʕir).
the-price PASS-raise.PST.3SG.M the-price
'The price was raised.' (past clauses in Jordanian Arabic (Jarrah 2023:15))

The contrast between (25) and (26) follows from the Arabic clausal structure and movement discrepancies between the past and the present tenses. According to Jarrah 2023, the underlying object in both tenses moves to the edge of vP/Voice (or AspP), yet since in the present the verb does not move to T, the postverbal order of the underlying object is disallowed.

Crucially, no such asymmetries hold in the present and past tenses in Sason Arabic. For example, the counterpart of Ferguson's *God wishes* are found in Sason Arabic as well, mostly with *ill-wishes* with unconventionalized expressions, with the VS order in both tenses.[12]

(27) a. i-qleb be-kki Allah!
3SG.M-destroy.PRS house-your.SG.F God
'May God destroy your home!'

b. qalab be-kki Allah!
destroy.PST.3M.SG house-your.SG.F God
'May God destroy your home!' (Sason Arabic, Ferguson's *God wishes*)

Turning to the other word order contrast from Jordanian Arabic (Jarrah 2023), we observe that preverbal and postverbal orders are licit in both tenses of SA. In SA, indefinite subjects tend to occur postverbally, but with equal possibility of appearing preverbally (Akkuş 2021a,b, 129). This holds both for present and past sentences. (28) demonstrates this for passive, and (29) for active clauses.

(28) a. (beyt-ma) ın-adal (beyt-ma).
house-a PASS.PST-build.PAST.3SG.M house-a
'A house was built.' (Akkuş 2021b:129,(254))

b. (beyt-ma) in-y-addel (beyt-ma).
house-a PASS.PRS-3SG.M-build.PRES house-a
'A house is built.' (passive clauses in Sason Arabic)

(29) a. (zıɣer-ma) ca (zıɣer-ma).
child-a come.PST.3SG.M child-a
'A child came.' (Akkuş 2021b:129,(253))

b. (zıɣer-ma) i-či (zıɣer-ma).
child-a 3SG.M-come.PRS child-a
'A child comes.' (active clauses in Sason Arabic)

The position of bare verbs with respect to adverbs and floating (subject) quantifiers can also be used to determine whether V remains within VP or raises to a higher functional category (Pollock 1989). Sentences in (30) show that adverbial adjuncts and floating quantifiers can appear between the verb and the direct object in SA past tense sentences.

(30) a. zıɣer kara *ams* maitub-ma.
child write.PST.3SG.M yesterday letter-a
'The child wrote a letter yesterday.'

b. zıɣar kar-o *kıll-en* maitub-ma.
children write.PST-3PL all-3PL letter-a
'The children all wrote a letter.' (verb position in the past in SA; Akkuş 2014:88,(12))

Same possibilities hold for present tense too, (31).

(31) a. zıɣer i-kri *lome* maitub-ma.
child 3SG.M-write.PRS today letter-a
'The child writes a letter today.'

b. zıɣar i-kr-o *kıll-en* maitub-ma.
children 3SG.M-write.PRS-PL all-3PL letter-a
'The children all write a letter.' (verb position in the present in SA)

Thus, it follows that the verb and the direct object in (30) and (31) are not within the same maximal projection. In other words, these sentences show that the verb has raised out of VP, over the adverb and the floating quantifier.[13]

Another systematic difference Benmamoun (2000) notes between present tense sentences and their past tense counterparts concerns whether the verb may merge with negation or not. For example, in Egyptian Arabic while the verb may not move to form a complex head with negation in the present, (32a), it must do so in the past, (32b).

(32) Egyptian Arabic negation (from Benmamoun 2000:4a-5a, taken from Jelinek 1981)

a. miš bi-yi-ktib.
NEG ASP-3SG.M-write
'He isn't writing.'

b. Omar ma-katab-š ig-gawaab.
Omar NEG-write.PST.3SG.M-NEG the-letter
'Omar didn't write the letter.'

The contrast between the present and past tenses in EA is taken to follow from the view that the verb in the past moves all the way to T head and a further step head-movement to the Negation head, which is located above T (Soltan 2007; Benmamoun et al. 2013). On the other hand, the verb may only move to a lower position in the present and merges with aspect morpheme, but does not have to merge with T since T lacks the [+V] feature.[14]

Turning to SA, we observe that unlike Egyptian Arabic, the verb must merge with the negation both in the present and past, (33).

(33) a. **mā**-ci-te.
NEG-come.PST-2SG.F
'You.(sg.f) didn't come.'

b. **mı**-tı-č-e.
NEG-2-come.PRS-F
'You.(sg.f) don't come.' (SA negation)

The discussion thus far establishes that the present tense behaves parallel to the past tense, which straightforwardly follows from the view that the present tense in Sason Arabic has undergone a change in the categorial features, ending up with the [+V, +D] paradigm, just like the past tense. Therefore, the emergence of a copula in the present verbless clauses with [+V, +D] features naturally follows as part of the shift in the categorial features in the language. Expectedly, the copula is negated by and merges with the same negative morpheme, *mā*, as in verbless clauses.

(34) a. ınte koys-e **mā**-kın-te.
you.SG.F beautiful-F NEG-be-2SG.F
'You.(sg.f) {aren't / weren't} beautiful.'

b. ınto wane **mā**-kın-to.
you.PL there NEG-be-2PL
'You.(pl) {aren't / weren't} there.' (SA)

The idea that the change in the present tense feature implicates a wholesale change in the copular constructions is also supported by the distributional differences of the copula in Arabic. In varieties such as Moroccan Arabic or Standard Arabic, while the copula is generally not available in the present tense, it is in some contexts, particularly with those with (generic) modal feature.

(35) a. Omar (*ta-y-kun) hna daba.
Omar (*asp-3sg.m-be) here now
'Omar is here now.'

b. Omar lazəm *(y-kun) hna
Omar necessary *(3sg.m-be) here
'Omar must be here' (MA)

Benmamoun (2000) suggests that such a contrast would follow from the assumption that the (generic) modal sentences contain a [+V] feature, as such the copula is inserted to check its categorial [+V] feature. On the other hand, other verbless sentences lack the [+V] feature, and hence the absence of the copula. Importantly, SA makes no such differentiation, and a copula is required in both types of clauses, (36), in line with all verbless clauses bearing the [+V] feature in SA.

(36) a. balki fı beyt *(kınt) sāhane.
maybe in house be.2sg.m now
'You may be at home now.'

b. fı beyt *(kınt) sāhane.
in house be.2sg.m now
'You are at home now.' (SA)

Moreover, the same copula is used in both tenses with event-in-progress interpretations of imperfectives in Sason Arabic, as shown in (37).

(37) a. sa *kınt-e* tı-fqız-e.
now be-2sg.f 2-run.prs-f
'You are running now.'

b. ams *kınt-e* kı-tı-fqız-e.
yesterday be-2sg.f asp-2-run.prs-f
'You were running yesterday.' (SA)

This is also in contrast to other Arabic varieties, e.g., Standard Arabic, which have the copula in past progressive, but not present progressive.

(38) a. ṭ-ṭaalibaat-u kun-na ya-ʔkul-na.
the-students.f-nom be.pst-3f.pl 3-eat-f.pl
'The students were eating.'

b. ta-ʔkulu ṭ-ṭaalibaat-u.
3f.sg-eat the-students.f-nom
'The students are eating.' (Standard Arabic)

The same property is observed in Egyptian Arabic too, it lacks copula in the present progressive, but has it in the past progressive.

(39) a. Fariid bi-yigmaʕ-hum.
Fariid IPFV-gather.3M.SG-them
'Farid is gathering them.'

b. Fariid kaan bi-yigmaʕ-hum.
Fariid be.PST.3M.SG IPFV-gather.3SG-them
'Farid was gathering them.' (Diesing and Jelinek 1995:22a-b)

Putting together the data thus far demonstrates that the present and past tenses behave parallel in SA in contrast to *non*-peripheral varieties. Therefore, an understanding of the SA tense system, particularly, the convergence of the present tense with the past tense by acquiring the [+V] feature explains a number of common properties in the language. Crucially, the [+V] feature also allowed for the development of the copula in the verbless sentences, available both in the present and the past.[15]

3.1.2 Further evidence from Sason Arabic against a null copula

Section 2.1 provided one of the arguments, based on Case, from Benmamoun (2000) that verbless clauses in Arabic varieties do not contain a null verbal copula in the present tense. In this section, we provide two more pieces of evidence from Sason Arabic against the null copula analysis.

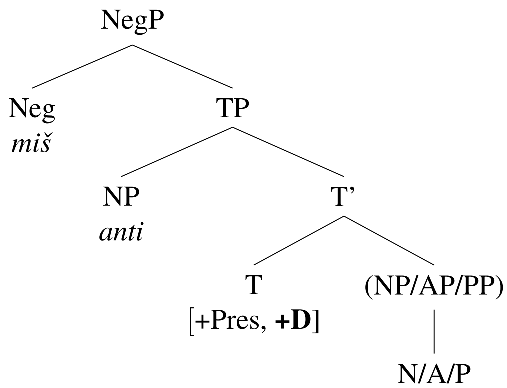
The first argument comes from the lack of negative pronouns in Sason Arabic. Many varieties such as Moroccan, Egyptian, Sana'ani Arabic have the so-called negative pronouns, in which negation merges with the subject pronoun. Consider (40).

(40) (Some) negative pronouns in Arabic (from Benmamoun and Al-Asbahi 2013:87)

Pronoun	**Sana'ani**	**Moroccan**
3m. sg.	miššu	mahuwaš 'not.3sg.m'
3f. sg.	mišši	mahiyaš 'not.3sg.f'
2m. sg.	mišant	mantaš 'not.you.sg.m'
2f. sg.	mišanti	mantiš 'not.you.sg.f'
1sg	mišana	manaši 'not.1sg'
1pl	mišeħne	maħnaš 'not.1pl'

Benmamoun and Al-Asbahi (2013) argue for example that the phenomenon in Sana'ani Arabic, where the pronominal subject follows the negation, is the result of negation merging with the subject pronoun in the present tense, which is located in Spec,TP, as schematized in (41) for 2f.sg negative pronoun *mišanti* (see also Benmamoun et al. 2013). They contend that the subject pronoun, which most likely encliticizes in the post-syntactic component, occurs in a context where it does not get pre-empted by a verb or tense that may merge with negation. Thus, the negative pronouns are enabled by the absence of an element on T head (see that work for a slightly different analysis for Moroccan Arabic).

(41)



Given this background, one expectation is that languages like Sason Arabic cannot have negative pronouns since the change in its categorial features of the present tense, it hosts an overt verbal copula there. This is indeed true, as shown by the ungrammaticality of (42) - same result holds for other persons as well.

(42) a. *ma-nana.
NEG-1PL.pron

b. *ma-[ı]nte.
NEG-2SG.F.pron

The second argument was briefly noted earlier, but it can be accentuated more: the idea that verbless sentences contain a verbal copula that is null raises the expectation that such a copula, if it can be overtly realized, should for all intents and purposes, follow the present tense paradigm in Arabic, differently from the past tense copula. Importantly, as has been seen in many examples thus far, Sason Arabic did not resort to a present tense form of the root KWN, instead the copula is (almost completely) homophonous in the present and past. We add another construction in which this identical form appears in both present and past tenses. Modal constructions also involve the same 'to be' copula across the board, as in (43).[16]

(43) a. balki fı beyt kınt sahane.
maybe in house be.2SG.M now
'You may be at home now.'

b. balki fı beyt kınt ams.
maybe in house be.2SG.M yesterday
'You might have been at home yesterday.'

In this section we provided several syntactic arguments to demonstrate that Sason Arabic has evolved a verbal copula in the present tense. Compared to the so-called non-peripheral varieties of Arabic, this is a major syntactic change in the Sason variety. If our analysis is on the right track, many important questions arise about how the verbal copula emerged. In the next section we take up this topic and we consider two scenarios, both present in modern Arabic varieties.

3.2 *Variation and change*

In section 3.1, we discussed how a change in the categorial features of the present tense in SA has also resulted in the emergence of a copula in verbless sentences. In this section, we briefly engage with the question of a possible historical trajectory that has resulted in the properties SA synchronically exhibits, including the emergence of a verbal copula as well as its position in the clause.

It is highly likely that SA developed a present tense with nominal and verbal features under contact with the neighboring languages Turkish, Kurdish or Armenian. Bilingualism and multilingualism are prevalent in the area where SA is spoken, though the penetration of the other languages and their impact depend on the density and size of the population, an important sociolinguistic issue that is beyond the scope of the paper, but one that deserves attention in order for us to understand the dynamics of language contact in the area and how it is impacting acquisition and change. Given the paucity of studies and the fact that SA does not have a written record that would have allowed us to examine the evolution of the development of the copula and other syntactic patterns, our remarks below must remain tentative and educated guesses for the moment.

3.2.1 *Word order in Sason Arabic copular clauses*

Now we turn briefly to an important property of the SA copula that seems to relate to language contact. The copular construction in SA exhibits head-final properties with a predicate+copula order, just like its neighboring languages Kurdish, Turkish, Zazaki and Armenian. This topic and other contact-induced syntactic changes are discussed in studies like Akkuş and Benmamoun 2016, 2018. For example, Akkuş and Benmamoun 2016 argue that the structure with verbless sentences is head-final with the copula in T.[17] Given the existence of prior studies dedicated to the influence of language contact, and that the primary interest of this paper, i.e., the featural change in the present tense, is dissociated from the order in which the copula can appear (see the cited sources for more discussion).

Unlike the verbal auxiliary, which necessarily precedes the participle, (44), the same copula in verbless clauses has to follow the predicate, especially with individual-level predicates, (45), such as *intelligent, smart, Arab*.

(44) a. ams kınt-e kı-tı-fqız-e.
yesterday be-2SG.F ASP-2-run.PRS-F
'You.(sg.f) were running yesterday.'

b. *ams kı-tı-fqız-e kınt-e.
yesterday ASP-2-run.PRS-F be-2SG.F
'You.(sg.f) were running yesterday.'

(45) a. ınte koys-e kınt-e.
you.SG.F beautiful-F be-2SG.F
'You.(sg.f) are/were beautiful.'

b. *ınte kınt-e koys-e.
you.SG.F be-2SG.F beautiful-F
'You.(sg.f) are/were beautiful.' (SA)

The head-final property of the copula in SA is in contrast with the Arabic varieties, which lack the overt copula, but can be shown via negative clauses bearing *laysa*, (46).

(46) Standard Arabic; Mohammad 2000:31,(105)

a. ʔal-walad-u {laysa ṭawīl-an}.
the-boy-NOM is.not tall-ACC
'The boy is not tall.'

b. *ʔal-walad-u {ṭawīl-an laysa}.
the-boy-NOM tall-ACC is.not
'The boy is not tall.'

It should be noted that in SA, the order of the copula is more flexible with stage-level predicates, in that while the post-predicate order is still the default order, displacing the copula to the pre-predicate position is allowed (with slight degradation for some speakers).

(47) a. ınte {cuan-e kınt-e}.
you.SG.F hungry-F be-2SG.F
'You.(sg.f) are/were hungry.'

b. ?ınte {kınt-e cuan-e}.
you.SG.F be-2SG.F hungry-F
'You.(sg.f) are/were hungry.' (SA)

The predicate-copula order is likely the result of contact with predominantly head-final languages such as Turkish, Kurdish, Armenian, what is pre-theoretically referred as a *calque* or areal feature. Our suspicion is that the reason why the copula construction is among the first to display head final order could be because there was no overt present tense copula in the language, so that would be a relatively easier entry point to adopt the head final order. As aforementioned, SA adopted head-final property in other domains such as light-verb constructions, (48) (Akkuş and Benmamoun 2018; Akkuş 2020a), which are also not a native construction to Arabic.[18]

(48) a. qazan sawa
win do.PST.3SG.M
'to win'

b. gerre/hās sawa
noise/sound do.PST.3SG.M
'to make noise/sound' (SA)

Thus, it is plausible that the head-final property is further making its way into the syntax of SA, applying to other constructions. Of course, only time will tell and it will all depend on whether this variety of Arabic survives long enough for the change in word order across constructions to play out. We still would like to reiterate that in principle a featural change in the present could take place without a change in the surface order of the copula, which we believe is the case in other Anatolian Arabic varieties such as Tillo/Siirt Arabic (Jastrow 2006; Talay 2007; Lahdo 2009), in which the copula is head-initial (see also fn. 4 for the same property in the light-verb constructions).

3.2.2 A potential historical path

In this section, we briefly touch upon a possible historical trajectory that may have resulted in the properties SA synchronically exhibits. As also noted above, this discussion is necessarily tentative for various reasons: firstly, like all diachronic research, we are limited by the available data, particularly for a language like Sason Arabic that does not have written records of any sort, and which would allow us to examine trends and patterns of change. Therefore, our discussion is a first approximation based on what is happening with neighboring Arabic varieties that are related to Sason Arabic, and on contact with its neighboring languages. Of course, the sociolinguistic history of population contact and diffusion which underlie them is a matter of considerable interest, and it would help shed light on the dialect differences we have uncovered. However, given the lack of such studies and resources, our focus has been on the theoretical implications. As such, this paper has refrained from speculating on the historical and sociolinguistic dynamics that may have impacted the syntactic evolution of the Anatolian Arabic varieties; instead our primary goal is to use the Sason Arabic data as empirical base to engage with the topic of variation in categorical feature specifications (Chomsky 1995) and their syntactic implications, which has merits in and of itself irrespective of how exactly the historical change has unfolded.

With the above important caveats in mind, we share below our thoughts on how the syntactic change may have played out, e.g., what parameter setting(s) may have changed. Adopting the framework of Roberts and Roussou 2003, we assume that language change consists of change in the realization/attraction property of functional heads, that is, a change in the lexicon. Similarly, and concurring with Lightfoot (1979, 1991) and Roberts and Roussou (2003, p. 11), we take it that syntactic change can be viewed as change in the parametric values specified for the functional heads of a given language.

In particular, in this approach, parameters are linked to a subclass of lexical items, that is, functional elements, which also have to be learned. It, then, follows that the same mechanism is also responsible for setting parameters. As such, while parameters are specified by UG, and cues are provided by the input, the parameters are set by the learning device on the basis of the interaction of cues and UG.

The parameter change could be about many types, e.g., the relative word order between an object and the verb, e.g., going from the predominant object-verb order in Latin to the verb-object order in modern Romance languages. It could also be, in our view and consistent with Roberts and Roussou 2003, about the categorial features of a particular functional head, which could in turn interact with another parameter. Applying this idea to Sason Arabic copula, the first parameter concerns the categorial specifications of the T head. Benmamoun (2000, p. 14) argues that (adopting Chomsky's (1995) categorial features), one can get the following feature structure of Tense.[19] In other words, within and across languages, functional categories can be parameterized as to what categorial features they contain, as represented in (49).

(49) *Categorial features*

	English	**French**	**most Arabic varieties**
Present	[+D, +V]	[+D, +V]	[+D]
Past	[+D, +V]	[+D, +V]	[+D, +V]

We suggest that the first step in the resulting change in SA involves a reanalysis of the featural specification of the present tense, which changed from [+D] to [+D, +V]. Given that the grammatical system - Sason Arabic - undergoing change exists side by side with one (or more) additional systems with different categorial feature setting, it is very likely that this change is triggered via language contact (most plausibly due to bilingualism). In particular, given that cues are provided by the input, the parameter regarding the categorial feature(s) on the present tense has been reset on the basis of cues from the languages SA has been in contact with. Following recent studies (e.g., Kupisch and Polinsky 2022; D'Alessandro et al. 2025), we maintain the view that the outcomes of language change in contact are overwhelmingly identical to language change in diachrony, so much so that the former can straightforwardly be predicted on the basis of the latter, and that language change under contact "can amplify and foreground developments that are known to take place in language diachrony" (Kupisch and Polinsky 2022:2), especially when functional features are the target of change. As such, syntactic change under contact can essentially be equated to accelerated diachronic change (Kupisch and Polinsky 2022).[20]

This change in the categorial features of the present tense in Sason Arabic yields the feature gain in (50), (in line with Roberts and Roussou 2003):

(50) *Categorial features in Sason Arabic*

Present	[+D, +V]
Past	[+D, +V]

Crucially, the reanalysis of categorial features on the present tense in SA from [+D] to [+D, +V] has resulted in another change to the attraction property of this functional head.[21] As such, another parameter, i.e., movement, has come to be associated with the present tense in Sason Arabic as well (see Roberts and Roussou (2003, ch. 3) for treating movement as a parameter).[22]

The addition of the [+V] feature to the present T which also changes its attraction property is consistent with both language internal properties, as has been demonstrated in section 3, and the properties of contact languages that provided the cues that interact with the parameters.[23] For example, both Iranian languages (including Persian, Kurdish and Zazaki varieties) and Turkish have been argued to involve V-to-T movement, as well as merger of the negation with the verbal complex in different tenses (see e.g., Ghomeshi 1996; Toosarvandani 2009; Rasekhi 2016; Akkuş 2020b; Akkuş et al. 2025; Osmani 2024 for some discussion of Iranian languages; and Kelepir 2001; İnce 2006; Gračanin-Yüksek and İşsever 2011 for Turkish).[24]

4 Conclusions

In this paper, we have investigated the syntax of verbless sentences in Arabic within the theory of categorial features and their syntactic role. Many Arabic varieties such as Moroccan Arabic lack a copula in the present tense while they have an overt copula in the past and future. On the other hand, understudied and so-called peripheral Arabic varieties such as Sason Arabic have an obligatory copula in both tenses.

Using the toolkit of the categorial features [+V], [+D] of Chomsky (1995), we have demonstrated that the presence or absence of a copula reveals a deeper property about the feature specifications of the present tense. In many Arabic varieties, the past tense has both [+V, +D] categorial features, the present tense is specified only for the [+D] feature even in sentences with verbal predicates. This asymmetry captures many morphosyntactic contrasts between past and present sentences. On the other hand, Sason Arabic has undergone a change in its present tense specification, such that it hosts both [+V, +D] categorial features, thus syntactically diverging from many varieties. The use of categorial features to account for intriguing and previously unaccounted variation in clause structure is mainly for heuristic purposes as far as this paper is concerned, but it also potentially could provide additional argument in support of non-interpretable (or uninterpretable) categorial features as drivers of a range of syntactic dependencies. This approach takes seriously the view that the functional domain of the clause drives syntactic dependencies (e.g., Borer 1984), and as such dialectal syntactic variation in some instances, likes the ones considered in this paper, may be the result of syntactic change in the functional domain.

As we mentioned at the outset, there is an ongoing debate about the status of syntactic features and particularly of categorial features. In addition to the question of whether they can be better grounded and motivated, another question that we have not addressed is whether the variation that we attributed to tense in Arabic varieties as far as its specification for the [+D] and [+V] feature, is only limited to the present tense and, if it is, whether there are syntactic or semantic reasons for that. We have not engaged that question partly because it is not within the main scope of the paper, which is to try to account for the presence of a copula in the present tense in Sason Arabic, but not in Egyptian Arabic and Moroccan Arabic, and partly because it is a question that we have not been able to investigate in depth but that we hope to do so in the future. It is indeed an intriguing question that deserves an answer, a question that the new data from Sason Arabic presented in this paper and the analysis it advanced have made even more acute.

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Notes

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1. The functional layer has a played a role in capturing patterns such as questions (movement to the complementizer domain), verb second phenomena (dependency between the tense domain and the complementizer domain), VSO order (movement of the verb beyond the internal subject).

2. Many studies still note the need for such categorial features, which are sometimes used under different labels, e.g., Haeberli 2003, Svenonius 2003, Rizzi 2013, Hill 2014, Panagiotidis 2015, Biberauer and Roberts 2015, Predolac 2017, Zeijlstra 2020, Larson and Samiian 2021. In fact, Adger 2003:ch. 2 makes a case for the necessity of categorial features.

We should also point out that we are not aware of any strong arguments against the [V] feature that led to the diminished role it has played in accounting for syntactic patterns that involve verbs and verb phrases. We suspect that the main reason that has to do with the debate about whether verb movement should belong in narrow syntax or should be relegated to the PF component. Chomsky (1995, 2001, 37-8) argued for the latter, but it is fair to say that the issue remains unsettled. Therefore, we continue to assume with Adger 2003; Zeijlstra 2020 that the [V] feature is part of the feature specifications of tense and is involved in driving the syntax of the verb and the verb phrase in narrow syntax. For example, Zeijlstra 2020, et seq. takes formal features to be (some sort of) categorial features, and connects them to the labeling algorithm. Even if turns out that verbal syntax belongs in the PF component that in itself would not be a strong argument against the syntactic nature of the [V] feature.

By the same token, even if it turns that the notion of categorial features should be abandoned or refined further, the data that we will discuss in depth below would still require an account that, in our review, would have to acknowledge some role for functional categories as factors in the observed variation.

3. We use the term 'future tense' mainly for expository purposes, noting (and following Abusch 1985, 1997 and many others) that future is not a simple tense but a complex tense composed of two parts. We also leave aside the discussion of copula in future tenses since unlike other Arabic varieties, SA uses the lexical verb 'to become' instead of a copula in copular sentences.

4. The copula is also required in imperatives and in the context of modals (Benmamoun 2000).

5. Some of the arguments discussed in Benmamoun 2000 besides Case, are based on complementizers, expletive subjects, temporal adverbs, among others, and provide support for a null tense node in the present tense. We assume with many others, including Fassi Fehri 1993 and Shlonsky 1997, that there is a present tense projection in the so-called verbless sentences in Arabic. Crucially, the presence of a tense node does not require a null copula.

6. Benmamoun (2000) provides additional independent arguments, synchronic and diachronic, to show that the present tense with verbal predicates is only specified for the [+D] feature.

7. See Benmamoun 2008 for various pieces of evidence as to why an analysis that relies on the morphophonological status of tense (i.e., whether a particular tense is an affix that must be supported by a lexical head or not, which would be the case for past tense in Arabic, but not present tense) along the lines of Baker 2003 cannot work for Arabic.

8. This group represents an older linguistic stratum of Mesopotamia against the *gələt* dialects. The terms *qəltu* vs. *gələt* dialects are due to Blanc (1964), who distinguished between the Arabic dialects spoken by three religious communities, Muslim, Jewish, and Christian, in Baghdad. He classified the

Jewish and Christian dialects as *qəltu* dialects and the Muslim dialect as a *gələt* dialect, on the basis of their word *qultu* 'I said' of Classical Arabic.

9. The difference is only limited to the third (singular) person in a way that relates to *individual*-versus *stage-level* distinction, thus we leave it aside for the most part. We also do not discuss 3sg since it uses PRON rather than the copula. Moreover, as seen in the table in (20), the 3pl has two copula forms in the past, and the present seems to have adopted only one of them. This also shows that the language has the [+/- past] tense distinction.

10. The adverbs *lome* 'today', *sāhane* 'now' are also compatible with the perfective form of the predicate, but they yield the expected past reference interpretations: 'earlier today', and 'just now', respectively.

11. The discussion in this section provides examples from a few Arabic varieties, which are meant to be representative, but certainly not exhaustive. For obvious reasons, we cannot possibly discuss all the Arabic varieties in such a short paper: first off, not every variety has been described and analyzed in sufficient detail to allow us to draw reliable conclusions. Secondly, the better-studied varieties that we do not include either align with the patterns we illustrate or do not exhibit them. That said, it is an empirical question how the point(s) and diagnostics we employ will hold for each and every Arabic variety. We hope future detailed research on individual varieties will shed light on possible further variation across Arabic dialects.

12. Another example, not shown here, is 'May God burn you!' which again exhibits VS order in both tenses. Note that while the VS order is the accepted configuration for younger generations, older generations maintain the contrast between the VS in the past and SV in the present.

13. Parallel restrictions are observed in compound tenses in the present (e.g., 'I am eating') and past (e.g., 'I was eating').

14. The verb, however, may optionally move to negation, which explains the optional discontinuous negative pattern we see in Egyptian Arabic in the present tense:

(i) ma-bi-yi-ktib-š.
NEG-ASP-3M-write-NEG
'He isn't writing.'

In Moroccan Arabic, the verb always merges with negation in both past and present tense. This is clearly independent of tense and its feature specification and may be a property of negative sentences, an important issue and contrast that we will put aside here.

15. As pointed out to us by Rajesh Bhatt (pers. comm.), the semantics of tense could also provide a partial answer to the absence of a copula in the present and its presence in the past in non-peripheral Arabic varieties (as well as other languages which encode tense systematically). The semantics of present tense is usually given a relatively simple treatment, and does not involve a semantic operator to derive the present interpretation. On the other hand, depending on which view of tense one adopts, past tense either requires a special pronoun with past presupposition, or an existential quantifier for anteriority. For example, eventive predicates are assumed to contain an event variable that must be bound by a modal or tense operator other than present tense (following Enç 1991). Accordingly, the extra piece/information in the past tense might need to be encoded, and 'to be' serves this purpose.

16. It is important to keep in mind that an overt verbal copula does surface in generic present tense sentences and in the context of modals in Standard Arabic, with the main verb in the imperfective paradigm (Benmamoun 2000, 2008).

17. For the purposes of this paper, we remain agnostic as to how this final order of the copula is derived, e.g., via predicate fronting via movement or whether verbless clauses are indeed head-final. There is an ongoing debate as to how different types of word order are derived, which is not the main focus of our paper and nothing in our discussion of categorial features hinges on it.

18. Crucially, the head-final property is the dominant pattern for younger generations, while elderly speakers mainly use the head-initial pattern, as such the light verb precedes the nonverbal element. Moreover, the age-group or generation in between tends to oscillate between the two orders. We interpret this to mean that although the emergence of the light verb construction is highly likely due to language contact (independently argued for by Versteegh 1997; Talay 2007; Lahdo 2009; Akkuş and Benmamoun 2018), the adoption of the head-final property seems to be gradually unfolding. Additionally, note that larger communities in Anatolia that have also calqued the light-verb construction from Turkish and Kurdish, still maintain the Arabic head-initial property (Talay 2007 for Mardin Arabic, and Lahdo 2009 for Tillo Arabic). We attribute this to the usual sociolinguistic or socio-political factors commonly affecting the degree of change.

19. See Benmamoun 2000 for a more comprehensive system that involves other tenses/moods and functional categories.

20. An important question concerns how distinct or how similar are the outcomes of syntactic/language change in diachrony (CID) and syntactic/language change in contact (CIC)? Although some linguists explicitly claim that CIC and CID, i.e., diachronic, endogenous change, basically proceed along the same lines (cf. Heine and Kuteva 2005), others highlight their differences (cf. Thomason and Kaufman 1988). This can indeed be seen in other types of changes in Sason Arabic, which presumably due to the role of language contact proceed at an accelerated pace, allowing an inter-generational comparison.

21. An anonymous reviewer asks whether the [+V] feature in the present tense of SA was 'triggered' by non-native speakers of SA. This is an intriguing perspective that deserves consideration, but it is highly unlikely that it applies to the current context for at least two reasons. First, given the sociolinguistic and socio-economic dynamics of the region, multilingualism has proceeded unidirectionally. While native speakers of SA have acquired other languages such as Kurdish, Armenian, and in the last five decades or so, Turkish (see Akkuş 2020a for discussion of the language contact), the opposite direction is almost non-existent because both the speakers of Kurdish and Turkish (being the regionally dominant and official language, respectively) have not needed to learn SA. Secondly, as noted in Akkuş 2020a, there is a significant number of speakers over the age of 60 or so, who are monolingual in SA, and still exhibit the patterns reported in this paper. In fact, the importance of the role and degree of contact is further illustrated by the inter-generational differences in various domains of the language. As noted in Talay 2001, older speakers tend to have pharyngealized consonants in Anatolian Arabic varieties, while younger generations lost them, most likely due to more contact with surrounding languages. In the domain of copula itself, older generations tend to actually show more classical patterns seen in other Arabic varieties. For example, as noted in fn. 4, older speakers how the same VS/SV contrast for the relevant idioms in SA, while younger generations no longer have that. See also fn. 4 for a similar path regarding the light verb construction.

22. Note that in Roberts and Roussou 2003, loss of movement is taken to be a structural simplification, a view we agree with. Still, if we consider parameters as creating a space of possible variation within which grammatical systems are distributed, the presence of systems that would involve less frequent and rarer parameters is not unexpected. The movement parameter in Sason Arabic can be considered to be of this type.

23. Note that the specification of tense for categorial features also predicts the existence of other types of change, in which a language may grammaticalize a different lexical item without changing the feature structure of the present tense. This is indeed the case in some varieties of Palestinian Arabic, where a new copula *baka* emerged that can be used interchangeably with *kaan* in all syntactic distributions. See Benmamoun and Shalash 2012 for discussion, including how it provides further evidence against a null copula analysis in the present tense.

24. For space reasons, we do not provide the relevant examples; the reader is referred to the cited sources.